Tianyun Tang

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Appointments

- **The University of Chicago** **Chicago, IL**
  Committee in Computational and Applied Mathematics
  William H. Kruskal Instructor 2025–present
- **National University of Singapore** **Singapore**
  Institute of Operations Research and Analytics
  Research Fellow (Advisor: Kim-Chuan Toh) 2024–2025

Education

- **National University of Singapore** **Singapore**
  Department of Mathematics
  Ph.D. in Mathematics (Advisor: Kim-Chuan Toh) 2020–2024
- **University of Science and Technology of China** **Hefei, China**
  School of Mathematical Sciences
  Bachelor of Science in Mathematics and Applied Mathematics 2016–2020

Recognitions

- Ivo and Renata Babuška Thesis Prize, American Mathematical Society, 2026
- Mathematical Programming Meritorious Service Award, Mathematical Optimization Society, 2025
- Louis Hsiao-Yun Chen Best Dissertation Prize, National University of Singapore, 2025
- SIAM Student Chapter Certificate, SIAM, 2024
- Graduate Tutor Commendation letters, National University of Singapore, 2023
- Best Graduate Researcher Award, National University of Singapore, 2023
- PhD Conference Award, National University of Singapore, 2023

Research interests

- Optimization, with an emphasis on semidefinite programming, low-rank optimization, and Riemannian optimization.
- Computational methods for high-dimensional probability distributions, including variational inference and optimal transport.

Preprints

- **Convex relaxations for the optimization of Markov processes**
  (with Hongyi Zhang and Yuehaw Khoo), arXiv:2607.09423 (2026), preprint.
- **A quadratically convergent alternating projection method for nonconvex sets**
  (with Nachuan Xiao, Shiwei Wang and Kim-Chuan Toh), arXiv:2511.22916 (2025), submitted.
- **Convex relaxation approaches for high-dimensional optimal transport**
  (with Yuehaw Khoo), arXiv:2511.13847 (2025), submitted.
- **A Bregman ADMM for Bethe variational problem**
  (with Yuehaw Khoo and Kim-Chuan Toh), arXiv:2502.04613 (2025), submitted.

Publications

- **A preconditioned augmented Lagrangian method for solving semidefinite programming problems**
  (with Kim-Chuan Toh), INFORMS Journal on Computing, accepted.
- **On the efficient computation of proximal operators of affine-constrained nonconvex functions**
  (with Di Hou, Kim-Chuan Toh and Shiwei Wang), Mathematics of Operations Research, accepted.
- **A low-rank augmented Lagrangian method for polyhedral-SDP and moment-SOS relaxations of polynomial optimization**
  (with Di Hou and Kim-Chuan Toh), Mathematical Programming, published online (2026). DOI: 10.1007/s10107-026-02395-5.
- **An exact penalty approach for equality constrained optimization over a convex set**
  (with Nachuan Xiao, Shiwei Wang and Kim-Chuan Toh), Mathematical Programming, published online (2026). DOI: 10.1007/s10107-026-02367-9.
- **Stiefel optimization is NP-hard**
  (with Zehua Lai and Lek-Heng Lim), Optimization Letters, 20(5) (2026), 1011–1023.
- **RiNNAL+: a Riemannian ALM solver for SDP-RLT relaxations of mixed-binary quadratic programs**
  (with Di Hou and Kim-Chuan Toh), Mathematical Programming Computation, 18(3) (2026), 679–735.
- **Exploring chordal sparsity in semidefinite programming with sparse plus low-rank data matrices**
  (with Kim-Chuan Toh), SIAM Journal on Optimization, 36(1) (2026), 90–119.
- **A low-rank augmented Lagrangian method for doubly nonnegative relaxations of mixed-binary quadratic programs**
  (with Di Hou and Kim-Chuan Toh), Operations Research, 74(3) (2026), 1626–1647.
- **Optimization over convex polyhedra via Hadamard parametrizations**
  (with Kim-Chuan Toh), Mathematical Programming, 214(1–2) (2025), 91–131.
- **Accelerating nuclear-norm regularized low-rank matrix optimization through Burer-Monteiro decomposition**
  (with Ching-Pei Lee, Ling Liang and Kim-Chuan Toh), Journal of Machine Learning Research, 25(379) (2024), 1–52.
- **A feasible method for general convex low-rank SDP problems**
  (with Kim-Chuan Toh), SIAM Journal on Optimization, 34(3) (2024), 2169–2200.
- **A Riemannian dimension-reduced second-order method with application in sensor network localization**
  (with Kim-Chuan Toh, Nachuan Xiao and Yinyu Ye), SIAM Journal on Scientific Computing, 46(3) (2024), A2025–A2046.
- **Self-adaptive ADMM for semi-strongly convex problems**
  (with Kim-Chuan Toh), Mathematical Programming Computation, 16(1) (2024), 113–150.
- **Solving graph equipartition SDPs on an algebraic variety**
  (with Kim-Chuan Toh), Mathematical Programming, 204(1–2) (2024), 299–347.
- **A feasible method for solving an SDP relaxation of the quadratic knapsack problem**
  (with Kim-Chuan Toh), Mathematics of Operations Research, 49(1) (2024), 19–39.

- **Minimizing cycles in tournaments and normalized q-norms**
(with Jie Ma), Combinatorial Theory, 2(3) (2022), #6.
- **Monochromatic subgraphs in iterated triangulations**
(with Jie Ma and Xingxing Yu), Electronic Journal of Combinatorics, 27(4) (2020), P4.18.

Teaching

As an instructor at the University of Chicago:

- STAT 24300: Numerical Linear Algebra (Autumn 2026)
- STAT 25100: Introduction to Mathematical Probability (Spring 2026, Winter 2026)

As a teaching assistant at the National University of Singapore:

- DSA2102: Essential Data Analytics Tools: Numerical Computation (Semester 1, AY 2023/24)
- MA3252: Linear and Network Optimization (Semester 2, AY 2022/23)
- DSA3102: Essential Data Analytics Tools: Convex Optimisation (Semester 1, AY 2022/23)
- MA1521: Calculus for Computing (Semester 2, AY 2021/22)

Referee for journals

- Operations Research
- Mathematical Programming
- SIAM Journal on Optimization
- Mathematics of Operations Research
- Mathematical Programming Computation
- IEEE Transactions on Signal Processing
- Journal of Optimization Theory and its Applications
- European Journal of Operational Research
- Journal of Global Optimization
- Acta Applicandae Mathematicae
- Journal of Computational and Applied Mathematics

Talks and presentations

- Stiefel Optimization is NP-hard, 27th Conference of the International Linear Algebra Society, 2026, Virginia Tech, Blacksburg, VA, U.S.
- A Bregman ADMM for Bethe variational problem, 2026 Midwest numerical Day, University of Notre Dame, IN, U.S.
- Convex relaxation approaches for high-dimensional optimal transport, 2026 Deep Learning meets PDE Workshop, University of Minnesota, Minneapolis, MN, U.S.
- Stiefel Optimization is NP-hard, 2026 AMS Sectional Meeting AMS Special Session, Boston College, Massachusetts, U.S.
- A Bregman ADMM for Bethe variational problem, 2026 Informs Optimization Society Conference, Atlanta, Georgia, U.S.
- Exploring chordal sparsity in semidefinite programming with sparse plus low-rank data matrices, 2025 International Conference on Continuous Optimization, Los Angeles, California, U.S.
- A feasible method for general convex low-rank SDP problems, the 25th International Symposium on Mathematical Programming, Montreal, Canada

- A feasible method for general convex low-rank SDP problems, 2024 INFORMS Optimization Society Conference, Rice University, Houston, Texas, U.S.
- A feasible method for solving an SDP relaxation of the quadratic knapsack problem, 2023 INFORMS Annual Meeting, Phoenix, Arizona, U.S.
- Solving graph equipartition SDPs on an algebraic variety, The 10th International Congress on Industrial and Applied Mathematics, Waseda University, Tokyo, Japan.
- Solving graph equipartition SDPs on an algebraic variety, 2023 SIAM Conference on Optimization, The Sheraton Grand Seattle, Seattle, Washington, U.S.